

IMPORTANT NOTICE: Independently created illustrative Quality Engineering case study, NOT an actual Emirates internal project or proprietary dataset.

1. CASE STUDY TITLE & CONTEXT

Title: End-to-End Aviation Operational Quality & Reliability
Context: Modern aviation networks rely on seamless operational synchronisation. This case study examines interconnected operational challenges across *Flight Operations* → *Ground Operations* → *Aircraft Turnaround* → *Baggage Handling* → *Passenger Service*. Variation or failures at one stage propagate downstream into operational and customer-quality issues.

2. PROBLEM STATEMENT

Hypothetical Scenario: Airline hub experiences variation across turnaround times, ground handling, baggage movement, and coordination. Objective: establish root causes, implement corrective improvements, and sustain quality without fabricated numbers.

4. PROCESS MAPPING & VARIATION POINTS



Note: Highlighted points represent potential quality variation nodes (timing drift, handoff friction, resource contention) – not actual Emirates failure points.

5. FISHBONE ANALYSIS (ISHIKAWA)

- Exploratory Areas:**
- **People:** Coordination & communication gaps.
 - **Process:** Sequencing & timing variation.
 - **Equipment:** GSE & hydraulic availability.
 - **Technology:** Real-time tracking latency.
 - **Materials:** Supply staging delays.
 - **Environment:** Weather & gate congestion.

7. PARETO ANALYSIS

Multiple Contributors → Pareto Analysis → Vital Few Critical Contributors (80/20 Rule). Focuses quality resources on significant systemic contributors instead of random variations.

3. SIPOC ANALYSIS

SUPPLIERS	INPUTS	PROCESS	OUTPUTS	CUSTOMER
Ground teams, maintenance, IT	Aircraft, pax, baggage, info	Arrival → Turnaround → Boarding	On-time departure, safe turnaround	Passengers & stakeholders

6. 5 WHY ROOT-CAUSE ANALYSIS

1. *Symptom:* Flight departure delayed during boarding.
2. *Immediate:* Transfer baggage reconciliation delayed.
3. *Process:* Late bag arrival from inbound flights.
4. *Coordination:* Inbound delay not synced with sorting.
5. *Underlying:* Lack of automated alert triggers.
6. **Root Cause:** Absence of integrated real-time feedback loop.

8. SYSTEM / PROCESS FMEA

RPN = Severity × Occurrence × Detection
Assesses broad network service risks and specific turnaround failure modes. (RPN is a risk-prioritisation aid, not sole decision basis).

9. COMPLETE ANALYTICAL CHAIN & PRIORITISATION



10. PDCA IMPROVEMENT

- **Plan:** Define operational problem & objective.
- **Do:** Pilot improvement on controlled basis.
- **Check:** Compare indicators before/after.
- **Act:** Standardise successful changes.

11. PILOT & VALIDATION

Tests change under controlled operational conditions. Assesses process consistency, reliability, customer impact, and sustainability without fabricating KPIs.

12. KPI / SPC MONITORING

Monitors turnaround consistency, process adherence, baggage quality, and early warnings. Illustrative monitoring dimensions only.

13. CONTROL & STANDARDISATION

Sustained via SOPs, process controls, defined responsibilities, reaction plans, periodic reviews, and continuous improvement.

14. COMPLETE QUALITY ENGINEERING METHODOLOGY FRAMEWORK

METHODOLOGY STAGE	PURPOSE & OPERATIONAL APPLICATION
SIPOC & Process Map	Define boundaries and visualise end-to-end operational workflow steps.
Fishbone & 5 Why	Explore multi-factor causes and drill down to underlying process mechanisms.
Pareto & FMEA	Prioritise vital contributors and assess operational failure risks (RPN).
PDCA, Pilot & Validation	Implement, test under controlled conditions, and verify improvement effectiveness.
KPI/SPC, Control & Standardisation	Monitor performance, establish robust controls, and drive continuous improvement.

15. POTENTIAL OUTCOMES

- Reduced operational variation & consistent turnarounds.
- Better coordination & reduced recurring failures.
- Stronger preventive quality culture & reliability.

16. KEY QUALITY ENGINEERING INSIGHT

"In aviation operations, quality is an end-to-end system. Reliable customer experience depends on controlling variation across interconnected processes."

DISCLAIMER: Illustrative case study independently developed by Jagriti Singh for the 100 Days TQM & Quality Engineering Challenge. Not an official Emirates project.